GREEN UNIVERSITY OF BANGLADESH

Department of Computer Science and Engineering

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

A Unified Digital Healthcare Platform
Connecting Patients, Doctors, and Pharmacies

Design Project I | CSE 308 | Section 242 D2

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Purpose of this document

This SRS describes the proposed functions, quality requirements, interfaces, constraints, and acceptance conditions for the first web-based version of the healthcare platform. It is intended to guide requirements review, system design, implementation, and testing. The document describes planned requirements; it does not claim that the features have already been implemented.

Scope note

The project will begin as a browser-based web application. A separate Flutter/mobile application, AI features, OCR, Docker-based deployment, and advanced external healthcare integrations are outside the current baseline unless the supervisor approves a later change.

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1. Introduction

1.1 Purpose

The purpose of this SRS is to define clear, testable requirements for a unified digital healthcare platform. It gives the project team and supervisor a shared understanding of the functions the first version is expected to provide.

1.2 Project Overview

The proposed system is a web-based platform that brings patients, doctors, hospitals, and pharmacies into one place. Patients can find doctors, request appointments, attend online consultations, access permitted medical reports, receive digital prescriptions, and search for medicines listed by participating pharmacies. Doctors and pharmacies will manage the information and services relevant to their roles.

1.3 Problem Statement

Healthcare information and services may be spread across separate providers. Patients can have difficulty finding a suitable doctor, arranging a consultation, keeping medical reports together, or checking medicine availability across pharmacies. The project aims to make these tasks easier through a shared online platform. The first version will depend on participating providers and the data they maintain; it will not represent every hospital or pharmacy in Bangladesh.

1.4 Objectives

• Provide one web interface for patients, doctors, hospitals, and pharmacies.

• Support doctor discovery and online appointment booking.

• Support browser-based video consultation between a patient and a doctor.

• Allow authorized users to store and access medical reports and digital prescriptions.

• Allow participating pharmacies to maintain medicine listings, prices, and stock information.

• Support medicine search, pharmacy comparison, and online order placement.

• Protect private information through authentication, role-based access, and appropriate validation.

1.5 Intended Audience

This document is intended for the project supervisor, student developers, testers, and anyone reviewing the system requirements. It may also be used as a reference when preparing design diagrams and test cases.

1.6 Definitions and References

SRS: Software Requirements Specification. **RBAC:** Role-Based Access Control. **EHR:** Electronic Health Record. **API:** Application Programming Interface. **WebRTC:** browser technology used for real-time audio/video communication. **PostGIS:** PostgreSQL extension for geographic data and location-based queries.

2. Overall Description

2.1 Product Perspective

The system will be a new web application with a frontend, backend API, relational database, and a video consultation service. The frontend will communicate with the backend through REST APIs. The backend will validate requests, enforce access rules, and read or update stored data. Video sessions will use WebRTC through a suitable service such as LiveKit.

2.2 Product Functions

• Account registration, login, logout, and role-specific access.

• Patient, doctor, hospital, and pharmacy profiles.

• Doctor search and profile viewing.

• Appointment requests, scheduling, status updates, and cancellation.

• Online consultation sessions for eligible appointments.

• Medical report upload, storage, and authorized viewing.

• Digital prescription creation and viewing by authorized users.

• Pharmacy medicine catalogue, price, and stock management.

• Medicine search and comparison across participating pharmacies.

• Order placement and basic order-status tracking.

• Location-based nearby-pharmacy search where location data is available.

• Notifications or status messages for important events.

• Administrative tools for managing accounts and platform records.

2.3 Operating Environment

Component	Planned environment
Client	Modern desktop or mobile web browser
Frontend	Next.js with TypeScript and Tailwind CSS
Backend	FastAPI (Python)
Database	PostgreSQL; PostGIS may be used for geographic searches
Authentication	Better Auth with role-based authorization; backend must validate permissions
Video	WebRTC using LiveKit or an equivalent configured service
Interfaces	REST APIs over HTTPS
Version control	Git
Deployment	Web hosting and managed PostgreSQL; final provider to be selected

2.4 Design and Implementation Principles

• The system should be modular so that patient, doctor, hospital, and pharmacy functions can be developed and tested separately.

• The interface should use clear labels and simple workflows.

• The backend, not only the frontend, must enforce access permissions.

• Only the minimum necessary personal and medical information should be exposed to each role.

• Medicine stock and price information should show that it is supplied by participating pharmacies and may change.

2.5 General Limitations

The usefulness of search, comparison, and order features depends on accurate data from participating providers. Video consultation depends on browser compatibility, network quality, and the configured video service. The platform is not intended to replace emergency medical services or professional clinical judgement.

3. User Classes and Roles

User role	Main permissions and responsibilities
Patient	Manage own profile; search doctors and pharmacies; book or cancel eligible appointments; join own consultations; view own permitted reports and prescriptions; search medicines; place and track orders.
Doctor	Manage professional profile and availability; view assigned appointments; conduct consultations; create and view prescriptions for authorized patients; access permitted patient records related to care.
Hospital / provider representative	Maintain the hospital or facility profile and relevant service information. Any appointment or doctor-management permission must be explicitly granted.
Pharmacy representative	Manage pharmacy profile; maintain medicine listings, prices, and stock; receive and update orders; view only order information needed to fulfil them.
Administrator	Manage platform accounts and roles, review reports of system issues, and maintain permitted platform-level records. Administrative access to medical content must be limited and auditable.

3.1 Access Control Rule

A user may perform only actions allowed for their assigned role and record ownership. For example, a patient must not be able to read another patient's medical report by changing an identifier in a URL or API request. The backend must check authorization on every protected request.

3.2 Account Verification

The project should define a verification process for doctors, hospitals, and pharmacies before enabling provider-specific functions. For the prototype, verification may be performed by an administrator using information supplied by the project team. The system must not imply that a provider is officially verified unless the verification process has actually been completed.

4. Functional Requirements

The word **shall** indicates a required behaviour for the planned system. Requirements marked as conditional depend on the relevant service or provider being configured.

4.1 Account and Authentication

ID	Requirement
FR-01	The system shall allow a user to register with the required account information.
FR-02	The system shall allow registered users to log in and log out securely.
FR-03	The system shall associate each account with an assigned role.
FR-04	The system shall restrict protected pages and API operations according to role and record ownership.
FR-05	The system shall allow users to view and update permitted profile fields.
FR-06	The system shall provide a way to recover access to an account, subject to the configured authentication service.

4.2 Doctor Discovery and Profiles

ID	Requirement
FR-07	The system shall allow patients to search available doctor profiles.
FR-08	The system shall allow doctor search results to be filtered by supported fields such as specialty or location.
FR-09	The system shall display available profile details such as name, specialty, qualifications supplied to the platform, consultation fee, and available appointment times.
FR-10	The system shall allow doctors or authorized administrators to update permitted doctor-profile information.

4.3 Appointments

ID	Requirement
FR-11	The system shall allow patients to request an appointment with a listed doctor.
FR-12	The system shall show available appointment slots based on the schedule recorded in the system.
FR-13	The system shall prevent two confirmed appointments from occupying the same doctor time slot.
FR-14	The system shall allow authorized users to view appointment details and status.
FR-15	The system shall allow patients to cancel an appointment subject to the platform's configured cancellation rules.
FR-16	The system shall allow doctors or authorized staff to confirm, reject, or update appointment status.

4.4 Video Consultation

ID	Requirement
FR-17	For an eligible appointment, the system shall provide a way for the patient and doctor to join a scheduled online consultation.
FR-18	The system shall restrict consultation access to the intended participants or other explicitly authorized users.
FR-19	The system shall display a useful message if the video service is unavailable or the user cannot join.
FR-20	The system shall not claim that video consultation is available until the required video service and session configuration are working.

4.5 Medical Reports and Records

ID	Requirement
FR-21	The system shall allow an authorized patient to upload or submit a medical report using supported file types and size limits.
FR-22	The system shall associate each report with the correct patient and relevant metadata, such as report type and date when supplied.
FR-23	The system shall allow patients to view their own stored reports.
FR-24	The system shall allow a doctor to access a patient's report only when the doctor has an authorized care-related relationship or explicit permission.
FR-25	The system shall validate uploaded files and deny unsupported file types.

4.6 Digital Prescriptions

ID	Requirement
FR-26	The system shall allow an authorized doctor to create a digital prescription for a patient under their care.
FR-27	A prescription shall include the patient, prescribing doctor, issue date, medicine details, dosage instructions, and other required fields supported by the design.
FR-28	The system shall allow the patient to view prescriptions issued to them.
FR-29	The system shall retain a record of prescription creation and subsequent permitted changes.
FR-30	The system shall restrict prescription access to authorized users.

4.7 Pharmacy Catalogue and Medicine Search

ID	Requirement
FR-31	The system shall allow an authorized pharmacy representative to create and update medicine listings.
FR-32	A medicine listing shall support fields such as medicine name, strength or form when available, price, stock status, and pharmacy.
FR-33	The system shall allow users to search medicine listings by supported search terms.
FR-34	The system shall compare available prices and stock information across participating pharmacies when matching listings are available.
FR-35	The system shall identify the pharmacy that supplied each listing and display the last-updated time where practical.
FR-36	The system shall support nearby-pharmacy search when pharmacy location data and user location or search area are available.

4.8 Orders

ID	Requirement
FR-37	The system shall allow a patient to place an order from an available pharmacy listing.
FR-38	Before accepting an order, the system shall validate the selected items and the stock information recorded by the system.
FR-39	The system shall store order items, quantities, prices at the time of ordering, pharmacy, and order status.
FR-40	The system shall allow patients to view their own order history and current order status.
FR-41	The system shall allow a pharmacy representative to view and update orders assigned to that pharmacy.
FR-42	If an order contains items supplied by more than one pharmacy, the system may split it into separate pharmacy orders. This feature must be implemented only if the project design and available data support it.

4.9 Notifications and Administration

ID	Requirement
FR-43	The system shall display relevant status updates for events such as appointment changes and order updates.
FR-44	The system should provide in-app notifications; email or SMS notifications depend on external service configuration.
FR-45	The administrator shall be able to manage accounts and role assignments according to defined permissions.
FR-46	The system shall record important security-sensitive actions, such as role changes and prescription updates, in an audit log where implemented.

5. External Interface Requirements

5.1 User Interface

• The application shall provide separate, role-appropriate dashboards or navigation for patients, doctors, hospitals, pharmacies, and administrators.

• Forms shall show required fields and clear validation messages.

• The interface shall adapt to common desktop and mobile browser screen sizes.

• Important actions such as booking, cancelling, placing an order, or saving a prescription shall show a success or error message.

• The interface should use readable text, consistent navigation, and clear status labels.

5.2 Software Interfaces

Interface	Requirement
Frontend to backend	The Next.js frontend shall communicate with the FastAPI backend through documented REST endpoints.
Database	The backend shall access PostgreSQL through a suitable database driver or data-access layer. Database credentials must not be exposed to the browser.
Authentication	The frontend may use Better Auth for authentication flows. The backend must independently verify the identity/session or trusted token and enforce permissions before returning protected data.
Video service	The system shall use WebRTC through LiveKit or an equivalent configured service for consultation sessions.
File storage	Medical files shall be stored using a storage method that supports access control. Public, guessable file links must not be used for private medical records.
Location search	PostGIS may be used to search for nearby pharmacies based on stored coordinates and an agreed distance or area.

5.3 Communication Interfaces

Production traffic shall use HTTPS. API requests and responses should use JSON unless a file upload or download requires another content type. Video and real-time communication will also depend on the protocols and network requirements of the selected video service.

5.4 File Upload Interface

The system shall define allowed file types and maximum upload sizes. It shall validate uploaded content, generate safe storage names, and avoid exposing internal storage paths. The exact allowed types and size limits must be set before implementation and testing.

6. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Security	All protected backend endpoints shall verify authentication and authorization; hiding a button in the frontend is not sufficient.
NFR-02	Privacy	Medical reports and prescriptions shall be accessible only to the patient and other explicitly authorized users.
NFR-03	Transport security	Deployed production traffic shall use HTTPS.
NFR-04	Credential safety	Passwords, secrets, database credentials, and service keys shall not be stored in public source code or sent to the browser.
NFR-05	Input validation	The backend shall validate and normalize user input before processing or storing it.
NFR-06	Reliability	The system shall handle expected errors, including invalid input, missing records, unavailable services, and failed requests, without exposing sensitive internal details.
NFR-07	Performance	For the prototype, common pages and ordinary database searches should normally respond within 3 seconds under the agreed test load, excluding video setup and large file transfer. The team shall record the test conditions.
NFR-08	Concurrency	The system shall prevent conflicting confirmed bookings for the same doctor and time slot, including when requests arrive close together.
NFR-09	Usability	Users shall be able to complete common tasks using clear labels and a consistent navigation structure.
NFR-10	Compatibility	The web application shall be tested on current versions of commonly used browsers, including Chrome and Firefox, and at least one mobile browser.
NFR-11	Maintainability	Frontend, backend, and database responsibilities shall be separated so that features can be updated with limited impact on unrelated modules.
NFR-12	Data integrity	The database shall enforce appropriate relationships and constraints for users, appointments, prescriptions, pharmacy listings, and orders.
NFR-13	Auditability	Security-sensitive actions shall be logged where appropriate, without unnecessarily copying medical details into logs.
NFR-14	Backup and recovery	The deployment plan shall define database backup and restoration procedures before real patient information is used.
NFR-15	Accessibility	Forms and core workflows should support keyboard navigation, visible focus, and labels that are understandable to users.

6.1 Security and Medical Information

This project handles sensitive information. Development and testing should use synthetic or consented test data, not real patient records without appropriate authorization and safeguards. Access control, secure file delivery, limited data collection, and careful handling of logs are required design concerns. The prototype must not be presented as clinically certified or legally compliant merely because these controls are planned.

7. Data Requirements

The following are the main logical data entities. Final field names, relationships, and database constraints will be defined during database design.

Entity	Example information	Important access rule
User	User ID, name, contact details, role, account status	A user may update only permitted account fields; role changes are restricted.
Patient Profile	Patient ID, date of birth if needed, contact and profile details	Visible only to the patient and authorized care providers.
Doctor Profile	Doctor ID, specialty, qualifications, fee, availability	Public profile fields may be searchable; private account fields remain protected.
Hospital / Facility	Facility ID, name, address, contact, services	Only authorized representatives may edit the facility record.
Pharmacy	Pharmacy ID, name, address, coordinates, contact	Only authorized pharmacy representatives may manage it.
Appointment	Appointment ID, patient, doctor, time, status, type	Visible to the patient, assigned doctor, and authorized staff.
Medical Report	Report ID, patient, file reference, type, date, metadata	Private by default; protected access required.
Prescription	Prescription ID, patient, doctor, issue date, medicine instructions, status	Accessible only to authorized users; changes should be traceable.
Medicine Listing	Listing ID, medicine name, strength/form, price, stock, pharmacy	Pharmacy can manage its own listings; public fields may be searchable.
Order and Order Item	Order ID, patient, pharmacy, items, quantities, prices, status	Patient sees own orders; pharmacy sees orders assigned to it.
Notification	Recipient, event type, message, read status, timestamp	Only the intended recipient may access it.
Audit Event	Actor, action type, target record, timestamp, result	Restricted to authorized review; avoid storing unnecessary sensitive content.

7.1 Data Integrity Rules

• Each main record shall have a unique identifier.

• An appointment must refer to an existing patient and doctor.

• A prescription must refer to an existing patient and authorized prescribing doctor.

• A pharmacy listing must belong to a pharmacy account or facility authorized to manage it.

• An order item shall preserve the price used when the order was placed, even if the listing price later changes.

• Deleting or deactivating an account must not silently break required historical records; the handling policy must be defined during design.

7.2 Data Retention

The project team shall define how long accounts, medical files, prescriptions, orders, and audit events are retained and how deletion requests are handled. For the academic prototype, test data should be used and retention rules should be documented before deployment.

8. Use Cases

8.1 Use Case UC-01: Book an Appointment

Field	Description
Primary actor	Patient
Preconditions	The patient is logged in; the doctor profile and available slots are listed.
Main flow	1. Patient searches for a doctor. 2. Patient opens the doctor's profile. 3. Patient selects an available slot and appointment type. 4. System validates the slot and patient details. 5. System creates the appointment and displays its status.
Alternative flow	If the slot is no longer available, the system rejects the request and asks the patient to select another slot.
Postcondition	An appointment record exists with the correct patient, doctor, time, and status.

8.2 Use Case UC-02: Conduct a Video Consultation

Field	Description
Primary actors	Patient and doctor
Preconditions	An eligible appointment exists and the video service is configured.
Main flow	1. Authorized participant opens the appointment. 2. System checks the participant's permission. 3. System provides the session entry point. 4. Participant joins using a supported browser. 5. Consultation proceeds through the configured video service.
Alternative flow	If the service or network is unavailable, the system displays an error and does not show the consultation as successfully connected.
Postcondition	The appointment remains linked to the consultation status where supported.

8.3 Use Case UC-03: Create a Digital Prescription

Field	Description
Primary actor	Doctor
Preconditions	The doctor is authenticated and authorized to access the patient for care.
Main flow	1. Doctor opens the authorized patient or appointment record. 2. Doctor enters medicine details and instructions. 3. System validates required fields. 4. Doctor submits the prescription. 5. System saves the prescription and makes it available to authorized users.
Alternative flow	If required information is missing or permission is denied, the system does not save the prescription and displays an appropriate message.
Postcondition	The prescription is stored and associated with the correct patient and doctor.

8.4 Use Case UC-04: Search and Order Medicine

Field	Description
Primary actor	Patient
Preconditions	Medicine listings from participating pharmacies are available.
Main flow	1. Patient searches for a medicine. 2. System displays matching listings with pharmacy, price, and recorded stock status. 3. Patient selects a listing and quantity. 4. System validates the listing and recorded stock. 5. Patient submits the order and receives an order reference and status.
Alternative flow	If the listing is unavailable or stock has changed, the system asks the patient to revise the order.
Postcondition	A valid order is stored and made available to the relevant pharmacy.

9. Constraints, Assumptions, and Dependencies

9.1 Constraints

• The initial deliverable is a web application; a separate native mobile application is not included.

• The system depends on information supplied by participating doctors, facilities, and pharmacies.

• Online video consultations depend on browser support, internet connectivity, and a correctly configured video service.

• Payment gateway integration is not a baseline requirement unless added and approved separately.

• The platform will not provide automated diagnosis, AI recommendations, OCR, or automated drug-interaction analysis in this version.

• Hospital bed/ICU availability and nationwide external hospital-system integrations are not baseline requirements.

• The system shall not claim that all pharmacies or hospitals in Bangladesh are connected.

9.2 Assumptions

• Users have access to a supported browser and an internet connection.

• Doctors and pharmacies entering data are responsible for keeping their profiles, schedules, medicine listings, prices, and stock information reasonably current.

• The project team can use test accounts and sample data during development.

• The supervisor will review major changes to scope and acceptance criteria.

9.3 Dependencies

• Availability of hosting, database, and secure file storage services.

• Successful configuration of authentication and role verification.

• Availability of a video service and its required credentials or infrastructure.

• Availability of pharmacy coordinates if nearby search is included.

9.4 Change Management

New features or changes to these requirements should be recorded with a short description, reason, affected requirements, and approval status. Changes that add substantial work should be reviewed with the supervisor before implementation.

10. Acceptance Criteria and Verification

The following checks can be used to confirm that the first version meets its main requirements. More detailed test cases should be prepared during implementation.

Test ID	Acceptance check	Expected result
AT-01	Register and log in using valid test accounts for different roles.	Each account can sign in and receives only the navigation and access appropriate to its role.
AT-02	Try to access another patient's report through a modified URL or API identifier.	The system denies access and does not return the report contents.
AT-03	Book an available doctor time slot.	An appointment is created and its status is shown.
AT-04	Attempt to book the same doctor time slot twice.	The system prevents conflicting confirmed appointments.
AT-05	Join a consultation as an authorized participant.	The participant can reach the configured video session when the service and network are working.
AT-06	Try to open a consultation as an unrelated user.	The system denies access.
AT-07	Upload a supported medical report file.	The file is accepted, linked to the correct patient, and visible only to authorized users.
AT-08	Upload a disallowed file type.	The system rejects the file and displays a clear message.
AT-09	Create a prescription as an authorized doctor.	The prescription is saved with the required patient, doctor, date, and medicine instructions.
AT-10	Search for a medicine listed by two participating pharmacies.	Matching listings show their pharmacy and available price/stock information.
AT-11	Place an order for an available listing.	The order is stored with items, quantity, price snapshot, pharmacy, and status.
AT-12	Try to update another pharmacy's listing.	The system denies the unauthorized change.
AT-13	Test core screens on desktop and mobile browser sizes.	The primary workflows remain readable and usable without major layout breakage.
AT-14	Simulate an API or video-service error.	The system shows a helpful error without exposing secrets or internal stack traces.

10.1 Verification Methods

• **Functional testing:** check each requirement using normal and invalid inputs.

• **Authorization testing:** verify role restrictions and record ownership at the API level.

• **Integration testing:** test frontend, backend, database, file storage, and video-service interactions.

• **Usability testing:** ask a small number of test users to complete key workflows and record problems.

• **Performance testing:** measure response times under a documented test setup.

10.2 Completion Conditions

The first version may be considered ready for academic demonstration when the agreed core workflows pass their acceptance checks, known limitations are documented, no critical access-control defect remains open, and the supervisor's required deliverables have been reviewed.

11. Out of Scope and Future Considerations

The following items are not part of the current baseline. They may be considered in a later version only after review and approval.

Item	Current status
Native Android/iOS or Flutter application	Out of scope; browser-based interface only.
AI diagnosis or AI-powered treatment recommendations	Out of scope.
OCR for reading prescriptions or medical reports	Out of scope.
Automated drug-interaction checking	Out of scope for the current baseline.
Payment gateway and automated refunds	Not included unless separately approved.
Nationwide integration with hospital systems	Out of scope; initial system uses participating providers or controlled sample data.
Hospital bed and ICU live availability	Not a baseline requirement.
FHIR or other external healthcare interoperability implementation	Not a baseline implementation requirement; may be studied later.
Automated prescription locking across unrelated providers	Not included in the baseline.
Advanced multi-pharmacy order optimization	Optional extension; not required for the initial release.

12. Glossary

Term	Meaning
API	A defined interface through which software components exchange data.
EHR	Electronic Health Record; a digital collection of a patient's health information.
RBAC	Role-Based Access Control; permissions are assigned according to user roles.
REST API	An API style commonly used for resource-based web communication.
WebRTC	Browser and application technology for real-time audio/video communication.
PostGIS	An extension for PostgreSQL that supports geographic data and location queries.
Acceptance criteria	Conditions that must be met for a feature or deliverable to be accepted.

End of document